

PROGRESS REPORT I

1. Dataset Description

A. Background and Context

The dataset used in this study comes from a Peruvian company that exports fresh ginger to international markets, primarily the United States and Europe. Unlike vertically integrated agricultural producers, this company does not cultivate its own ginger. Instead, it sources raw product from independent farmers located in different regions of Peru.

Because the company competes with other buyers for supply, operational efficiency is critical. One key competitive advantage is the ability to process incoming ginger quickly and pay farmers promptly. Currently, the full processing cycle takes approximately seven days on average, which may reduce supplier loyalty and create incentives for farmers to sell to competitors offering faster payment. This operational challenge motivates the present analysis.

B. Processing Workflow

Once the ginger arrives at the facility, it undergoes a structured processing pipeline consisting of several stages:

a. Reception and Quality Inspection

The quality control department evaluates the incoming product and records the percentage of various diseases and defects, including:

- ✓ Warty disease
- ✓ Black foot
- ✓ White rot
- ✓ Mechanical damage

These measures provide an assessment of initial product condition.

b. Initial Classification

The product is sorted into three categories:

- ✓ Exportable product
- ✓ Domestic product
- ✓ Rejected material (including sand and unusable components)

c. Curing Process

Exportable ginger is placed in a controlled environment where it undergoes a drying process known as *curing*. During this stage, the product loses weight primarily due to water evaporation. The curing period typically lasts approximately three days.

d. Post-Curing Reclassification

After curing, the product is inspected again because diseases may develop or worsen during storage.

e. Final Weighing and Packaging

The exportable product is weighed and packaged immediately after the final classification. This weight determines the effective quantity received by the company and the amount paid to farmers.

C. Study Objective

The primary objective of this project is to predict the percentage of weight lost during the curing process using information available at the time the product arrives at the factory.

Specifically, the goal is to estimate curing-related water loss based on:

- ✓ Initial quality indicators
- ✓ Processing characteristics
- ✓ Initial weight measurements

Accurate predictions would enable the company to:

- ✓ Improve payment planning
- ✓ optimize processing schedules
- ✓ reduce delays
- ✓ strengthen supplier relationships
- ✓ enhance operational competitiveness

D. Dataset Structure

The dataset contains batch-level observations, where each row represents a lot of ginger received from farmers. The key variables used in this analysis include:

Response Variable

- ✓ Cured (%): Percentage of weight loss during the curing process (primarily water evaporation)

Processing Variable

- ✓ Days to process: Number of days between reception and final packaging

Quality Indicators

- ✓ Warty disease (%)
- ✓ Black foot (%)
- ✓ White rot (%)
- ✓ Mechanical damage (%)

Weight Measures

- ✓ Gross packed weight (kg)
- ✓ Received weight (kg)

Temporal Variable

- ✓ Receiving date

E. Relevance for Predictive Modeling

This dataset is particularly suitable for statistical learning because it combines:

- ✓ Biological variability (disease presence)
- ✓ Operational factors (processing time)
- ✓ Physical characteristics (weight measures)
- ✓ Temporal effects (receiving dates)

Understanding how these variables influence curing losses provides actionable insights into process optimization and supply chain management.

2. Data Exploration and Data Pre-processing

A. Create New Features:

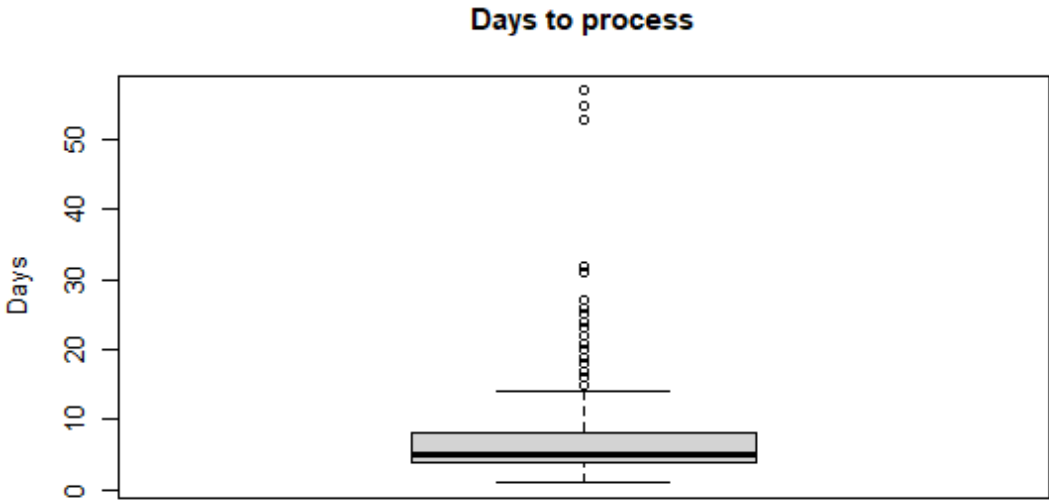
To better understand the operational performance of the processing system, a new variable called Days to Process was created. This feature measures the time required for each batch of ginger to move through the company's processing pipeline.

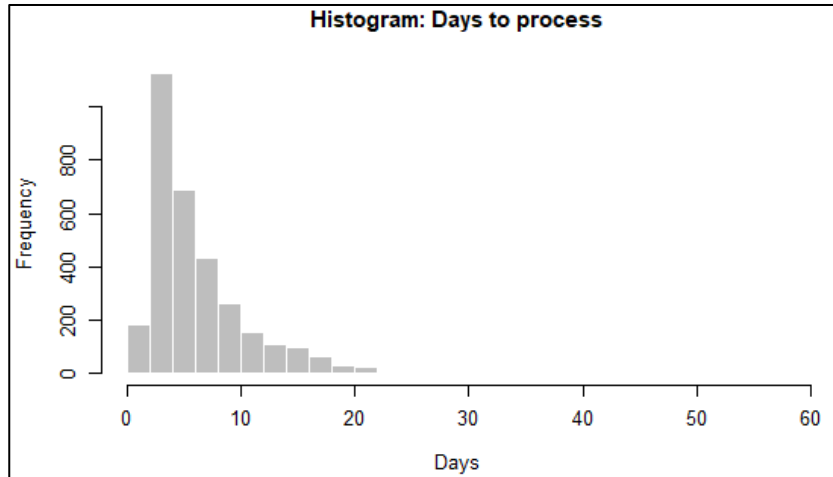
The variable was computed as the difference between the product's receiving date and the final packing date.

B. Removing rows

Exploratory analysis of the **Days to Process** variable was conducted using summary statistics, a boxplot, and a histogram. The results revealed a strongly right-skewed distribution with a median of approximately 5 days and a mean of 6.69 days. Several extreme values were observed, including processing times exceeding 50 days, which are operationally unlikely and may represent exceptional delays or data entry errors. The boxplot confirmed the presence of multiple observations beyond the upper whisker, indicating potential outliers. To reduce the influence of these extreme values, observations falling outside 1.5 times the interquartile range (IQR) were removed. This cleaning step improves the robustness of subsequent modeling by ensuring that the analysis focuses on typical operational conditions while minimizing distortion caused by rare or anomalous processing events.

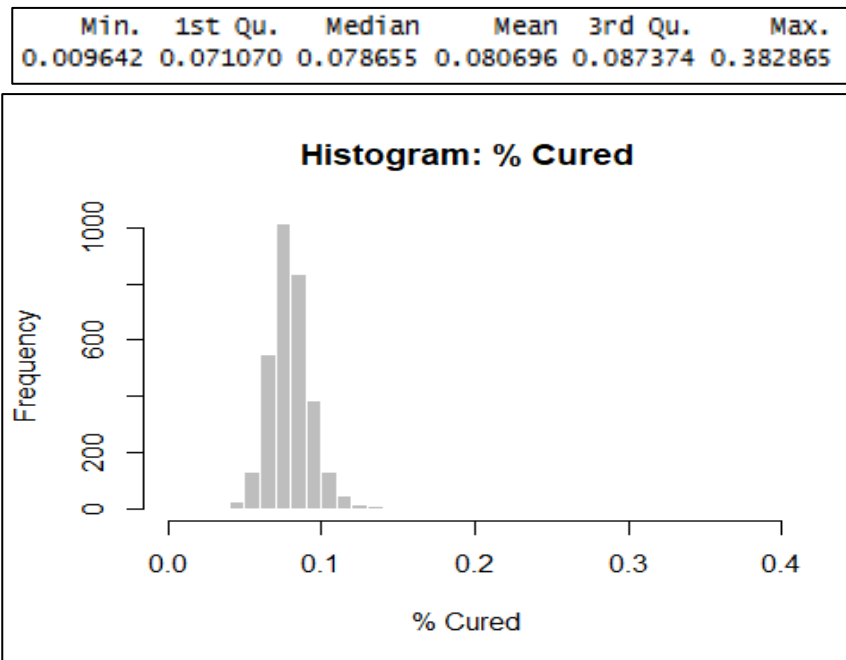
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1.00	4.00	5.00	6.69	8.00	57.00





Exploratory analysis of the **Cured (%)** variable was conducted using summary statistics and a histogram. The results indicate that curing losses are concentrated within a relatively narrow range, with a median of approximately 0.079 and a mean of 0.081. Most observations fall between 0.071 and 0.087, suggesting that the curing process is generally stable under normal operating conditions.

However, the distribution exhibits a noticeable right tail, with extreme values reaching approximately 0.38. These unusually high curing percentages likely reflect atypical processing conditions, measurement errors, or rare operational disruptions that do not represent the typical behavior of the system.

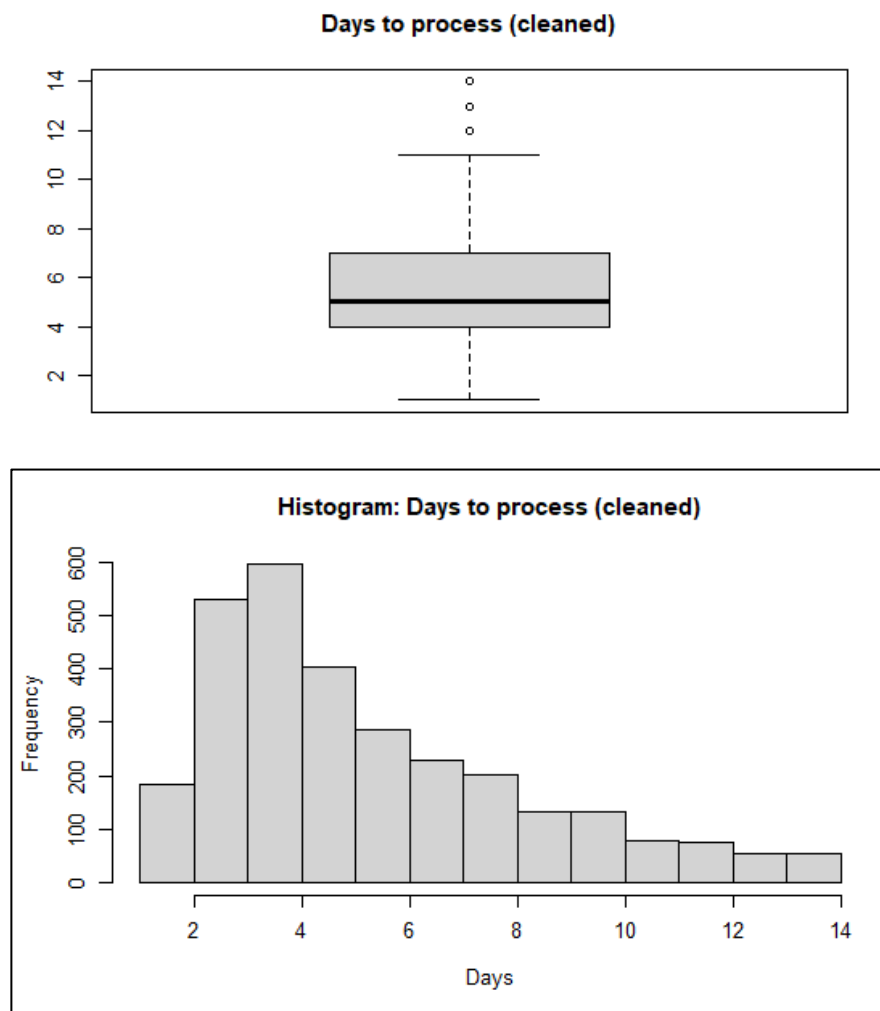


3. Data Visualization

After removing the rows with outliers, we can see that the mean of the Days to process is approximately 6 days.

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1.00	4.00	5.00	5.76	7.00	14.00

In the boxplot, we still observe some outliers, but they much less compared to the previous boxplot.



4. Comments

The preliminary analysis of the dataset focused on understanding the operational characteristics of the ginger processing system and preparing the data for predictive modeling. A key step in this process was the creation of the **Days to Process** variable, which quantifies the time required for each batch to move from reception to final packaging. This feature provides a meaningful operational measure of efficiency and directly relates to the company's objective of reducing payment delays to farmers.

Exploratory analysis using summary statistics, histograms, and boxplots revealed that both Days to Process and Cured (%) exhibited right-skewed distributions with the presence of extreme values. In the case of processing time, unusually large observations suggested atypical operational delays that are unlikely to represent standard production conditions. Similarly, extreme curing percentages reflected rare events that could distort the relationship between product characteristics and weight loss.

To ensure a robust and representative dataset, observations exceeding 1.5 times the interquartile range were removed. This step reduced the influence of anomalous values while preserving the inherent variability of the process. The resulting distributions better reflect typical operational behavior, characterized by moderate processing times and relatively stable curing outcomes.

From a practical perspective, the exploratory analysis highlights that the curing process operates within a controlled and predictable range, yet remains sensitive to operational and quality-related factors. This balance between stability and variability supports the feasibility of modeling curing-related weight loss using information available at the time of reception. Overall, the data preparation and exploratory steps establish a solid foundation for subsequent analysis by ensuring that the dataset accurately represents the company's operational reality while minimizing the impact of rare or exceptional events. This structured approach enhances both the interpretability and reliability of the predictive modeling phase.

APPENDIX

#To extract the data

```
library(readxl)
```

```
data <- read_excel("C:/Users/carlo/Documents/SFSU/4th Semester/Math  
748/Project/DataBase.xlsx", sheet = "DB")
```

Convert percent columns

```
pct_cols <- c("Cured (%)", "Warty disease (%)", "Black foot (%)",  
             "White rot (%)", "Mechanical damage (%)")
```

```
for (col in pct_cols) {  
  data[[col]] <- as.numeric(sub("%", "", data[[col]]))  
}
```

make sure numeric columns are numeric

```
data$`Days to process` <- as.numeric(data$`Days to process`)  
data$`Gross packed weight (kg)` <- as.numeric(data$`Gross packed weight (kg)`)  
data$`Received weight (kg)` <- as.numeric(data$`Received weight (kg)`)
```

```
head(data)
```

```
#Initial plots
```

```
summary(data$`Days to process`)
```

```
sum(is.na(data$`Days to process`)) #Analyzing missing values
```

```
boxplot(data$`Days to process`,main="Days to process", ylab="Days")
```

```
hist(data$`Days to process`,breaks = 20,main = "Histogram: Days to process",  
      xlab = "Days", col = "gray", border = "white")
```

```
summary(data$`Cured (%)`)
```

```
hist(data$`Cured (%)`,breaks = 30,main = "Histogram: % Cured",  
      xlab = "% Cured", col = "gray", border = "white")
```

```
#Removing outliers
```

```
Q1 <- quantile(data$`Days to process`, 0.25, na.rm=TRUE)
```

```
Q3 <- quantile(data$`Days to process`, 0.75, na.rm=TRUE)
```

```
IQRv <- IQR(data$`Days to process`, na.rm=TRUE)
```

```
lower <- Q1 - 1.5*IQRv
```

```
upper <- Q3 + 1.5*IQRv
```

```
clean_data <- data[data$`Days to process` >= lower &  
  data$`Days to process` <= upper,]
```

```
summary(clean_data$`Days to process`)
```

```
boxplot(clean_data$`Days to process`,  
  main="Days to process (cleaned)")
```

```
hist(clean_data$`Days to process`,  
  main="Histogram: Days to process (cleaned)",  
  xlab="Days")
```

